

WESTOVER AFB, MA, USA

WMO: 744910

Lat: 42.200N

Lon: 72.533W

Elev: 74

StdP: 100.45

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14703

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-15.4	-24.2	0.4	-16.1	-21.6	0.6	-13.0	12.0	-2.5	10.9	-3.4	2.0	40	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	32.6	23.0	31.0	22.0	29.5	21.2	24.5	29.9	23.7	28.5	22.9	27.1	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.9	26.5	22.3	17.2	25.9	21.5	16.3	25.3	74.2	29.9	71.0	28.4	67.9	27.2	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	7.7	-24.9	35.7	3.6	1.6	-27.5	36.8	-29.6	37.8	-31.6	38.6	-34.3	39.8		
(5)			-25.1	26.2	3.6	0.8	-27.7	26.8	-29.8	27.3	-31.8	27.7	-34.4	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-4.2	-3.1	2.2	8.8	14.8	19.4	22.8	21.3	17.2	11.1	4.8	-0.4
(7)		DBStd	10.33	6.06	6.15	5.33	4.81	4.32	3.42	2.94	2.92	4.14	4.41	4.81	5.21
(8)		HDD10.0	1673	440	368	247	79	7	0	0	0	39	169	324	
(9)		HDD18.3	3538	698	601	499	289	129	27	2	7	71	229	406	581
(10)		CDD10.0	1536	1	1	6	42	154	282	397	349	218	72	13	2
(11)		CDD18.3	360	0	0	0	2	19	58	141	98	38	3	0	0
(12)		CDH23.3	3571	0	0	3	49	264	616	1388	899	327	23	0	0
(13)		CDH26.7	1184	0	0	0	14	86	186	531	272	93	2	0	0
(14)	Wind	WSAvg	3.0	3.4	3.6	3.8	3.6	3.0	2.6	2.4	2.2	2.4	2.8	3.1	3.2
(15)	Precipitation	PrecAvg	1176	80	72	98	92	95	110	105	114	112	109	86	100
(16)		PrecMax	1659	167	236	177	190	178	256	255	253	264	350	147	174
(17)		PrecMin	813	34	35	19	25	27	43	50	30	30	13	16	21
(18)		PrecStd	208	30	39	40	41	37	58	55	59	70	71	30	35
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.4	16.1	21.7	28.8	32.1	33.7	35.4	33.3	32.3	25.9	20.4	16.6	
(20)		0.4% MCWB	13.6	11.9	13.5	17.8	20.7	23.2	24.7	23.7	23.2	18.7	14.8	14.2	
(21)		2% DB	10.2	11.3	16.7	23.8	29.2	30.8	32.9	31.1	29.3	23.0	17.7	12.8	
(22)		2% MCWB	8.1	6.6	10.1	14.4	19.7	21.5	23.3	22.4	21.4	17.4	14.1	10.6	
(23)		5% DB	6.6	8.0	13.4	20.6	26.3	28.6	31.5	29.6	27.0	21.0	15.5	10.1	
(24)		5% MCWB	4.5	4.6	8.0	12.6	17.5	20.0	22.5	21.5	20.4	16.9	12.0	7.9	
(25)		10% DB	4.0	5.3	10.6	17.7	23.7	26.9	29.8	28.0	24.9	19.0	13.0	7.5	
(26)		10% MCWB	1.6	2.6	6.6	11.4	16.0	19.2	21.6	20.9	19.3	14.8	9.2	5.5	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	13.8	13.4	15.2	18.4	22.9	24.5	25.8	25.2	24.4	21.5	17.2	14.7	
(28)		0.4% MCDB	14.5	15.5	19.8	24.4	29.8	31.1	32.3	30.6	29.4	23.3	18.5	15.4	
(29)		2% WB	8.5	8.0	11.7	16.8	21.0	23.4	24.8	24.0	23.0	19.5	15.0	11.3	
(30)		2% MCDB	9.0	10.2	15.0	21.7	26.1	28.7	30.5	28.3	26.8	21.5	16.8	12.6	
(31)		5% WB	4.5	4.9	8.9	14.4	19.6	22.2	24.0	23.1	21.8	17.7	12.5	8.2	
(32)		5% MCDB	6.4	7.3	12.3	18.4	23.7	26.3	28.9	27.0	25.4	20.3	15.1	9.6	
(33)		10% WB	2.0	2.8	6.7	12.1	17.8	20.9	23.1	22.2	20.6	15.6	10.0	5.6	
(34)		10% MCDB	3.6	5.0	10.1	17.1	22.0	24.6	27.4	26.2	23.6	18.0	12.5	7.2	
(35)	Mean Daily Temperature Range	MDBR	9.8	10.9	11.0	12.8	13.2	12.4	12.4	12.7	12.8	12.2	10.7	9.0	
(36)		5% DB MCDBR	11.2	13.7	16.0	18.3	17.7	15.1	14.3	14.2	14.5	14.5	13.5	11.9	
(37)		5% WB MCWBR	9.8	10.1	9.7	9.6	8.5	7.1	5.8	6.5	7.4	8.8	9.8	10.1	
(38)		5% WB MCDBR	10.8	12.4	13.6	14.7	14.4	12.4	11.9	11.5	11.7	11.4	11.2	11.0	
(39)		5% WB MCWBR	10.0	9.9	9.2	9.4	7.9	6.7	5.4	5.8	6.6	8.1	9.4	10.6	
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.306	0.331	0.373	0.410	0.457	0.474	0.443	0.381	0.341	0.311	0.293	
(41)		taud	2.441	2.411	2.431	2.357	2.289	2.189	2.150	2.224	2.375	2.501	2.543	2.527	
(42)		Ebn at Noon	858	900	916	895	866	823	804	818	849	850	833	830	
(43)		Edh at Noon	76	92	102	118	130	144	148	134	107	84	69	65	
(44)	All-Sky Solar Radiation	RadAvg	1.78	2.63	3.66	4.67	5.18	5.61	5.85	5.23	4.28	2.78	1.94	1.45	
(45)		RadStd	0.14	0.23	0.26	0.50	0.56	0.55	0.35	0.28	0.32	0.28	0.17	0.15	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+49

Nomenclature: See separate page